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(54) **HAPTIC FEEDBACK GLOVE**

(71) Applicant: **AxonVR Corporation**, Seattle, WA (US)

(72) Inventors: **Jacob A. Rubin**, Seattle, WA (US); **Robert S. Crockett**, San Luis Obispo, CA (US); **Marc Y. Goupil**, San Luis Obispo, CA (US); **Jeffrey D. D'Amelio**, San Luis Obispo, CA (US); **Bodin L. Rojanachaichanin**, San Luis Obispo, CA (US); **Kurt C. Sjoberg**, San Luis Obispo, CA (US); **Paul Piller**, San Luis Obispo, CA (US); **Nicholas J. Bonafede, JR.**, San Diego, CA (US)

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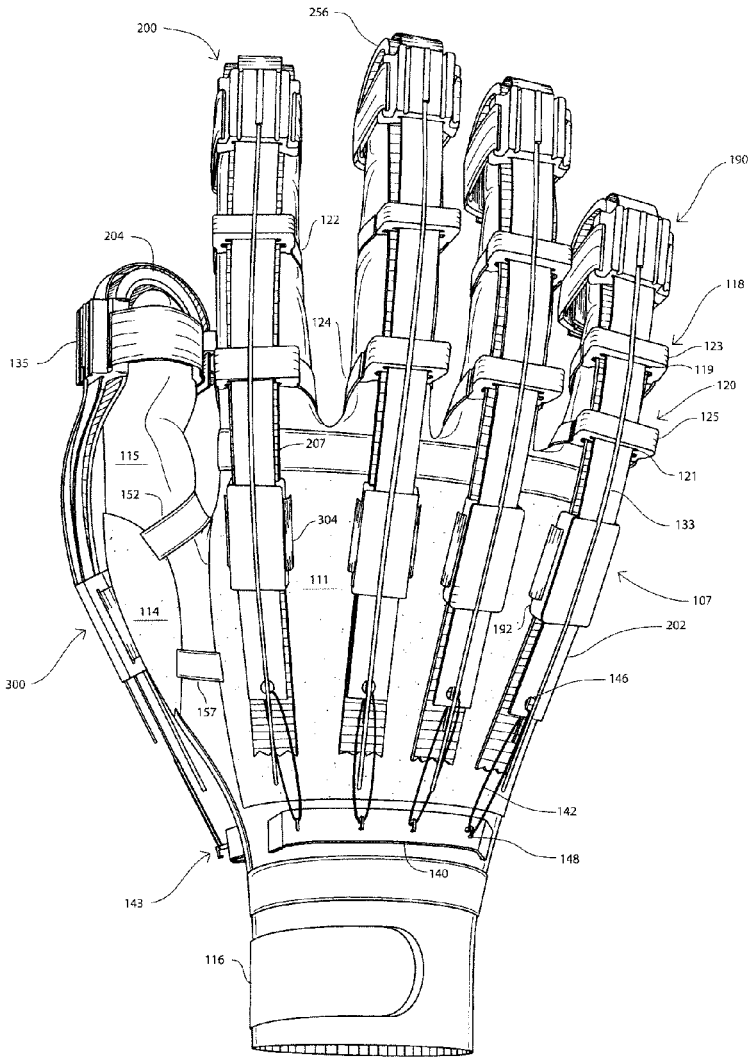
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(57) **ABSTRACT**

A human-computer interface system including: a sensor configured to transduce the location of a finger of a hand of a user; an exoskeleton configured to exchange mechanical energy with the finger; and an interface garment, including: an interface laminate coupled to a counterpressure assembly, configured to stimulate the user by applying a pressure to the finger; a first plurality of tactile actuators configured to apply a pressure to a finger of the user; and a second plurality of tactile actuators configured to apply a pressure to a palm of the hand of the user.



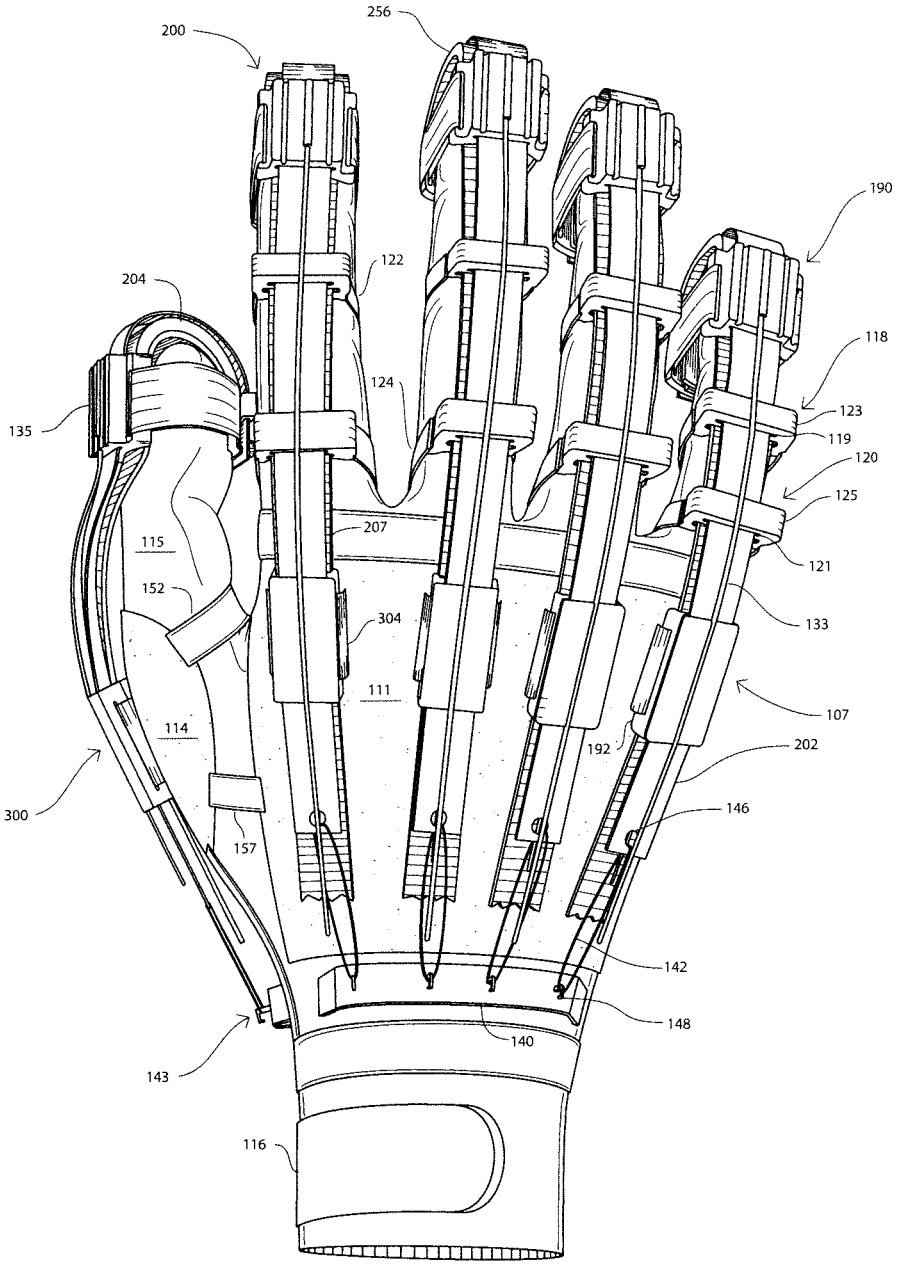

200
256
190
204
122
118
124
123
135
119
120
125
207
115
121
152
133
304
111
107
114
192
202
300
146
157
142
148
140
143
116

FIG. 1a

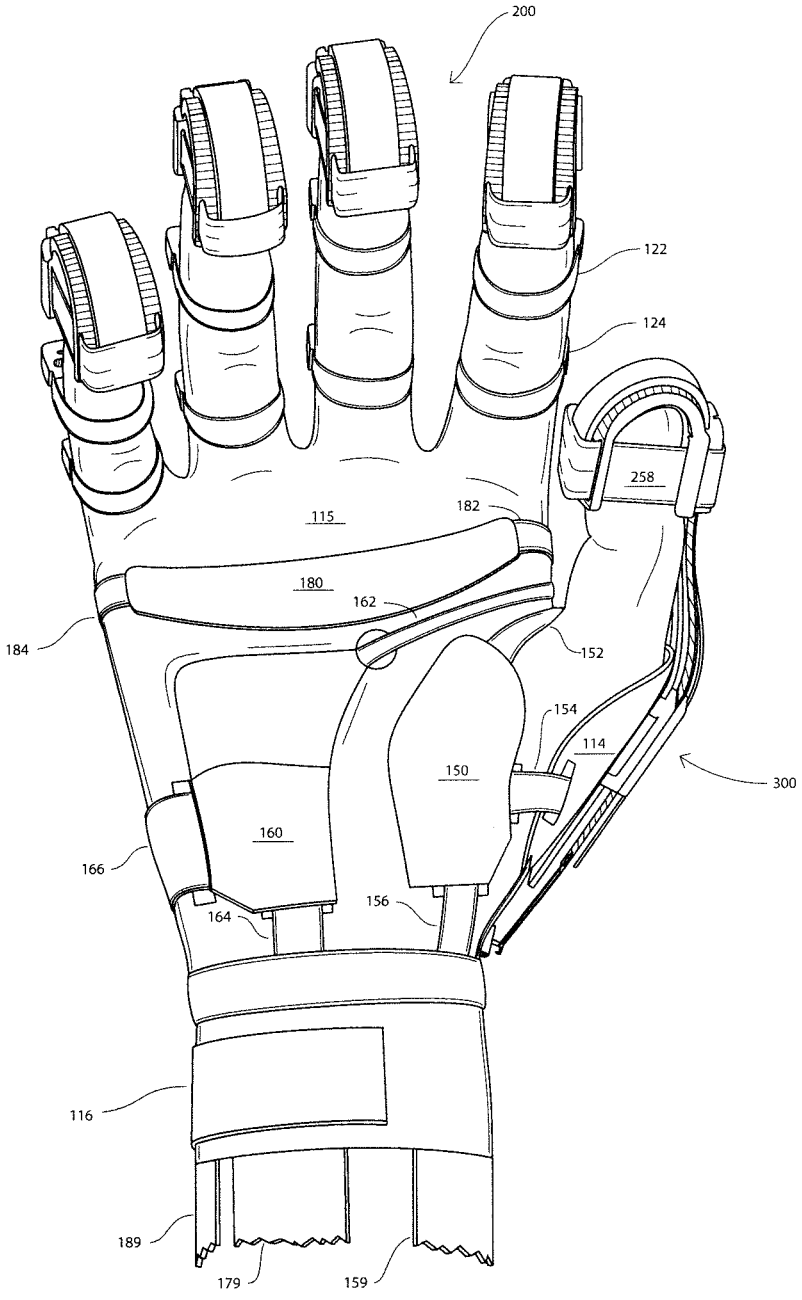

200
122
124
258
115
182
180
162
184
152
154
114
150
300
160
166
156
164
116
189
179
159

FIG. 1b

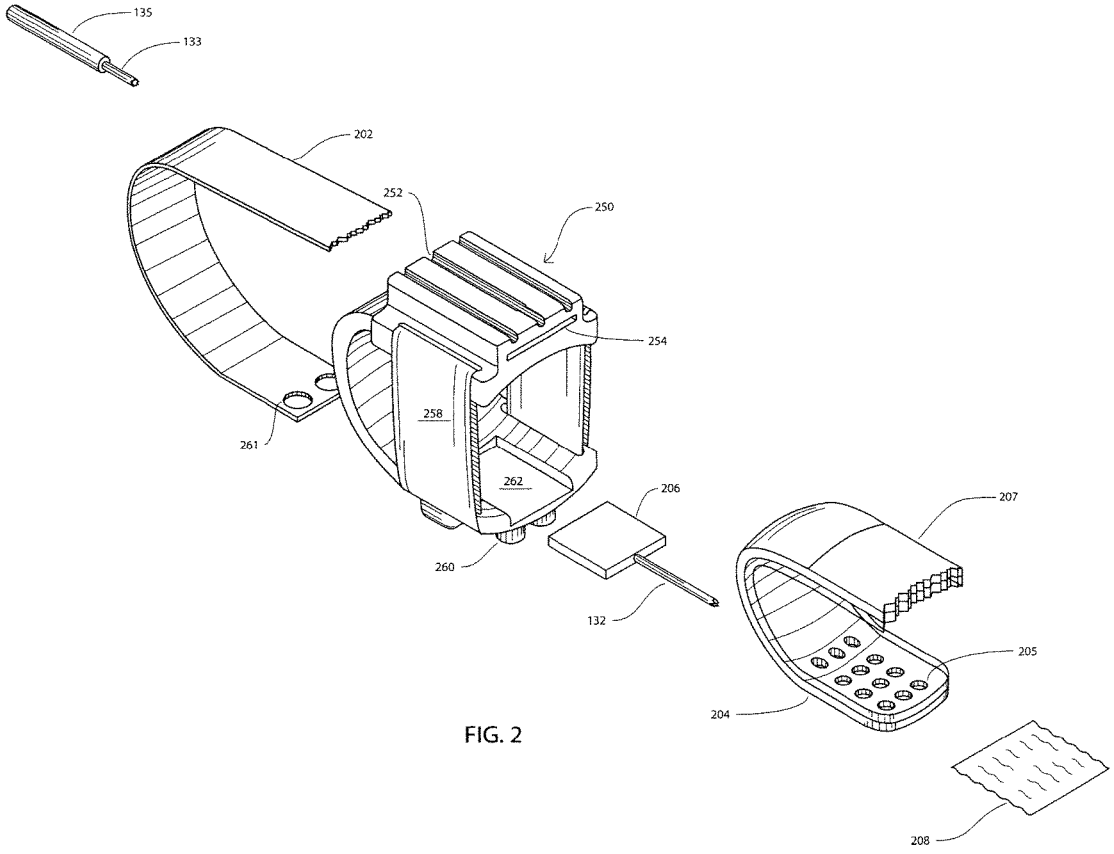


FIG. 2

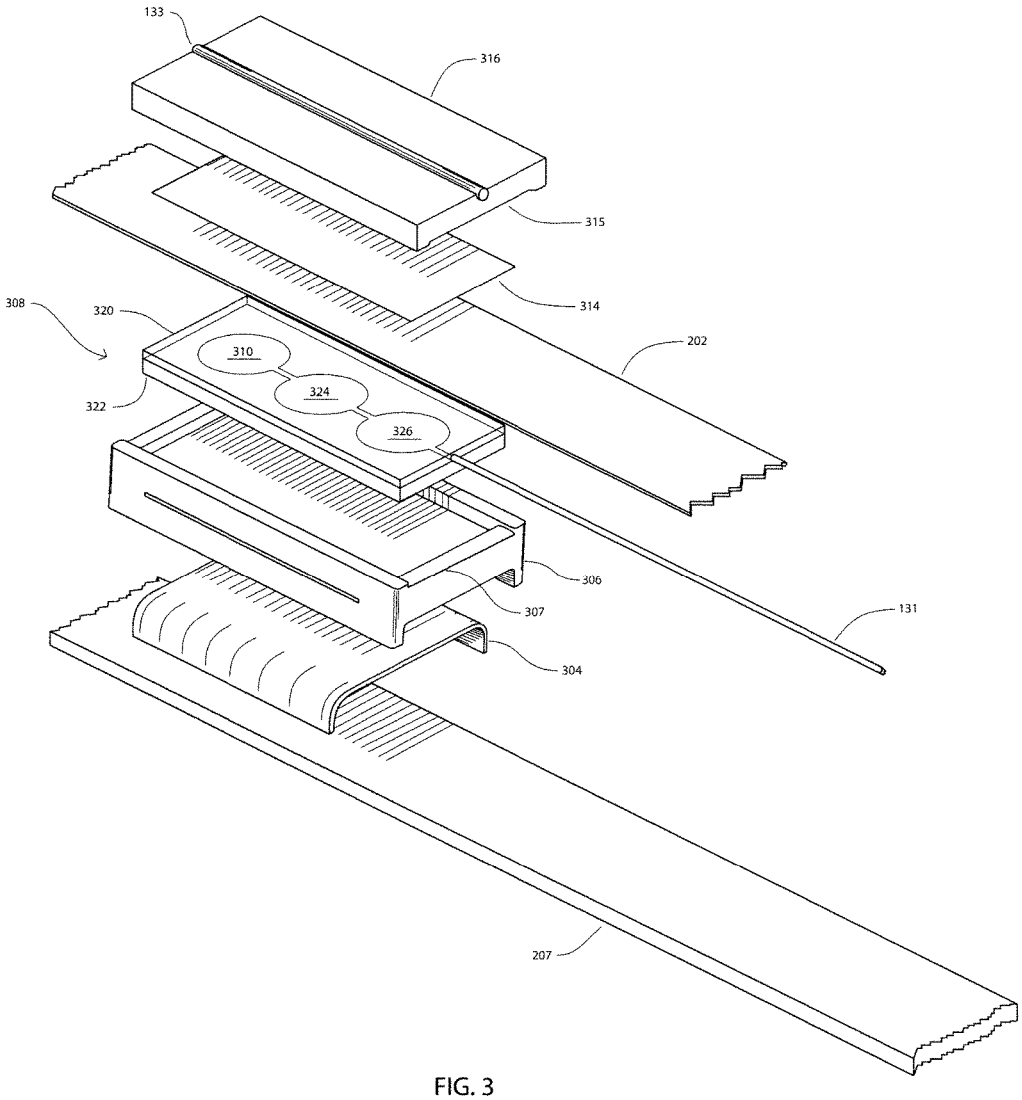


FIG. 3

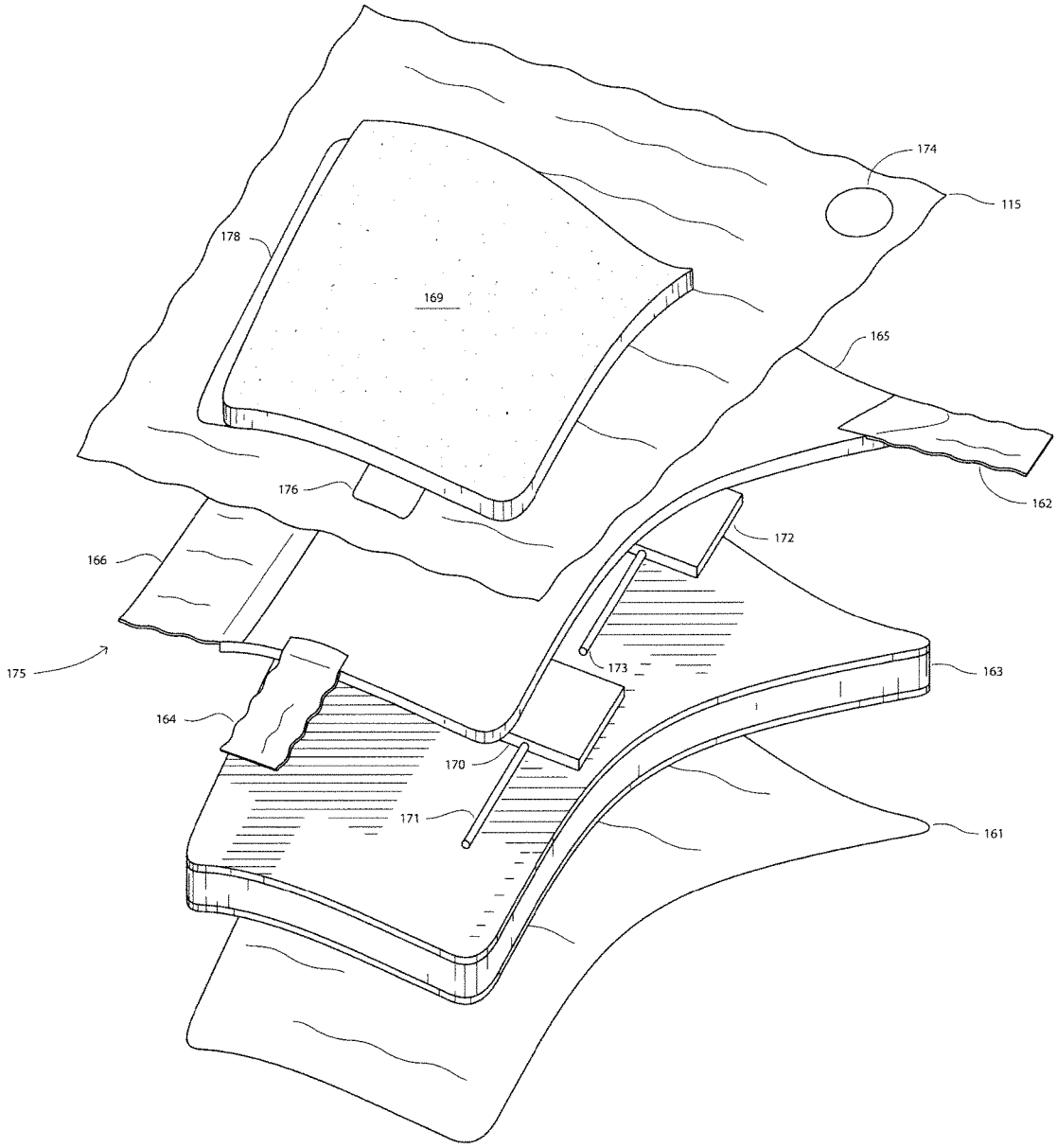


FIG. 4

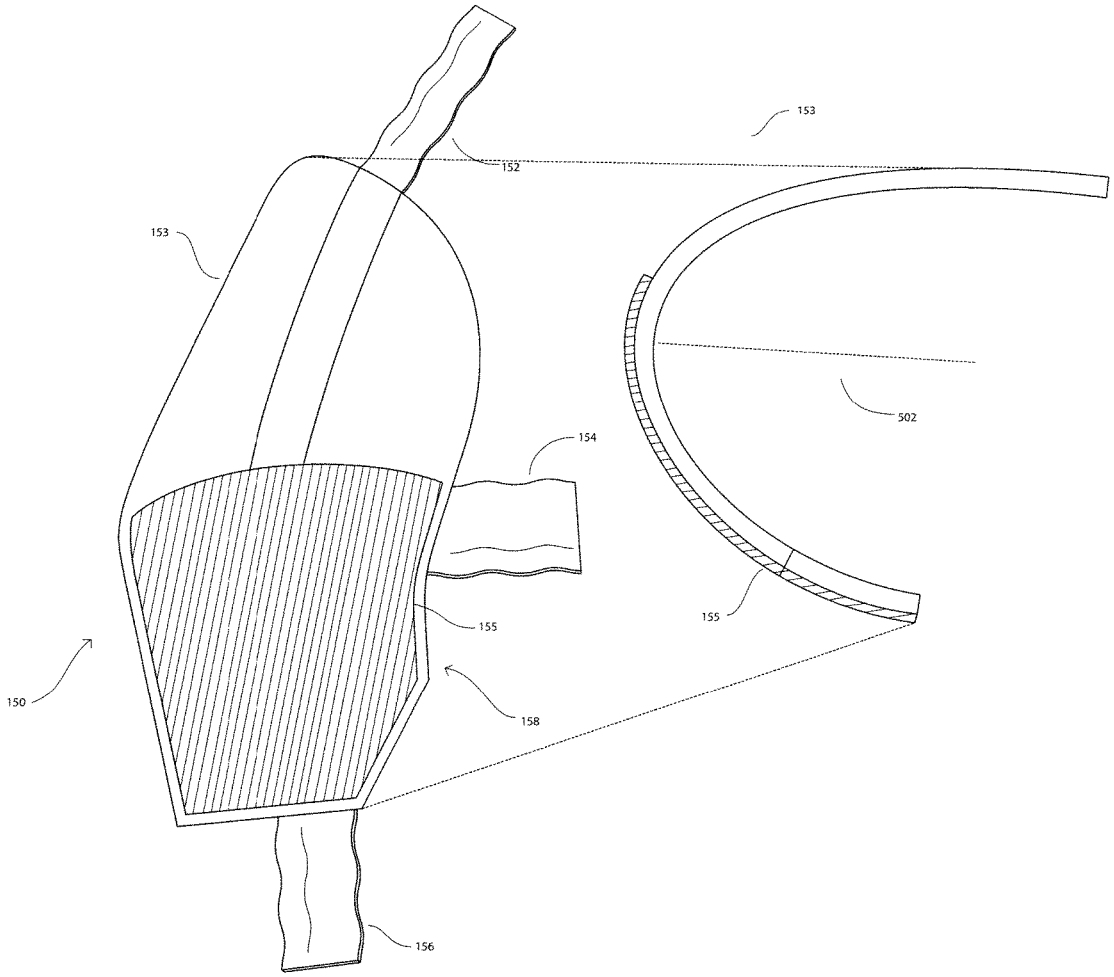


FIG. 5

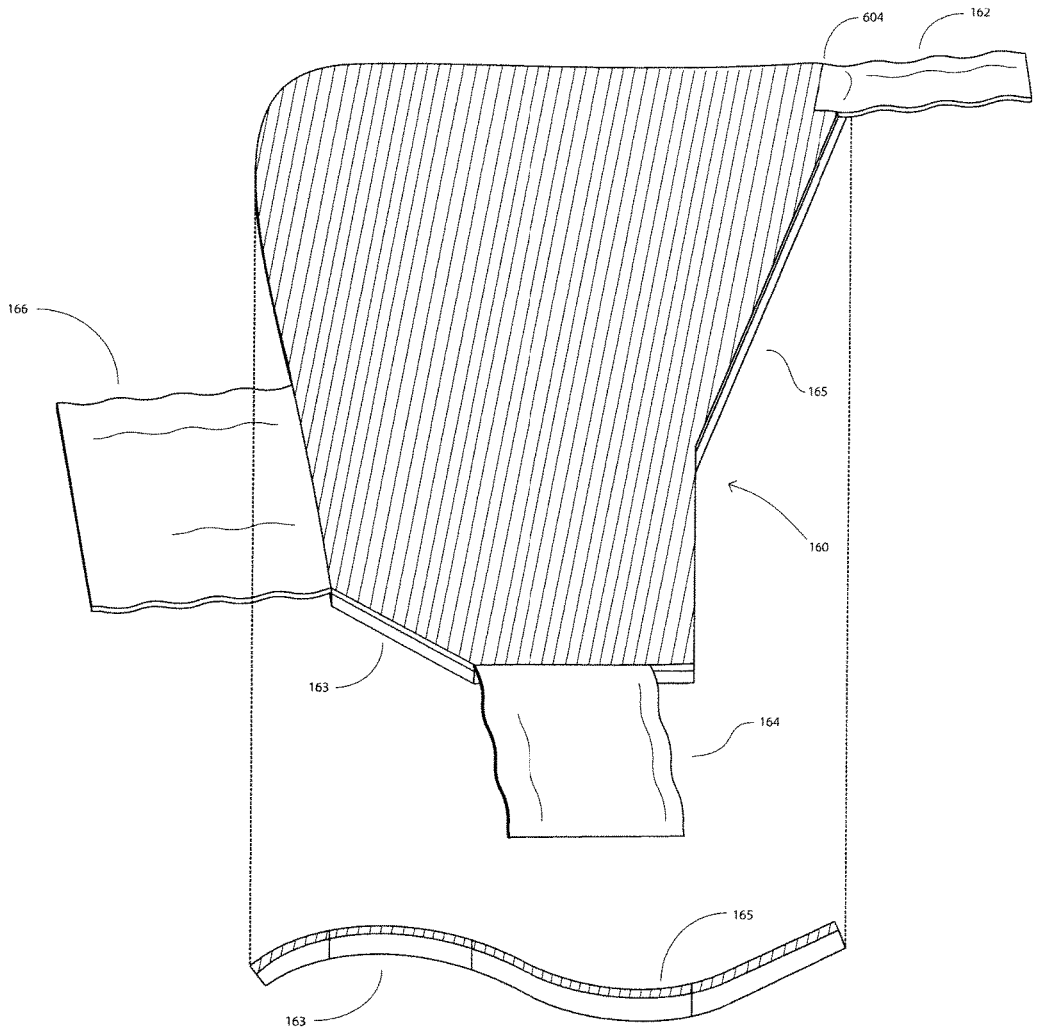


FIG. 6

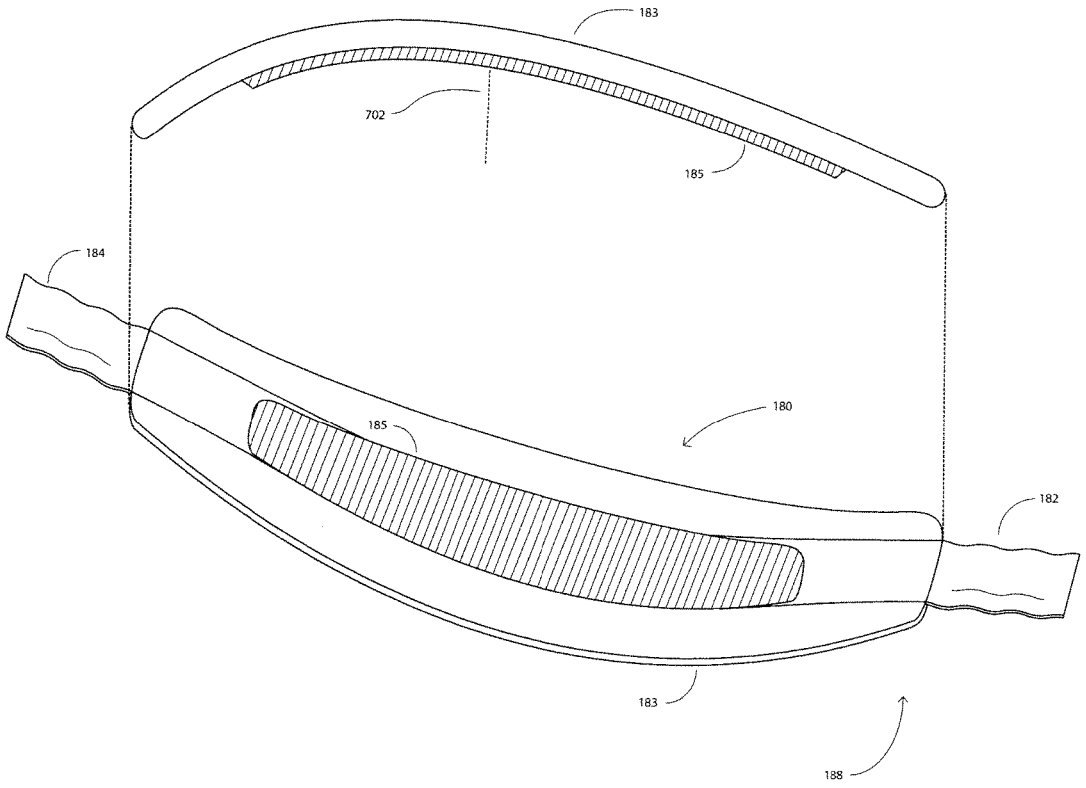


FIG. 7

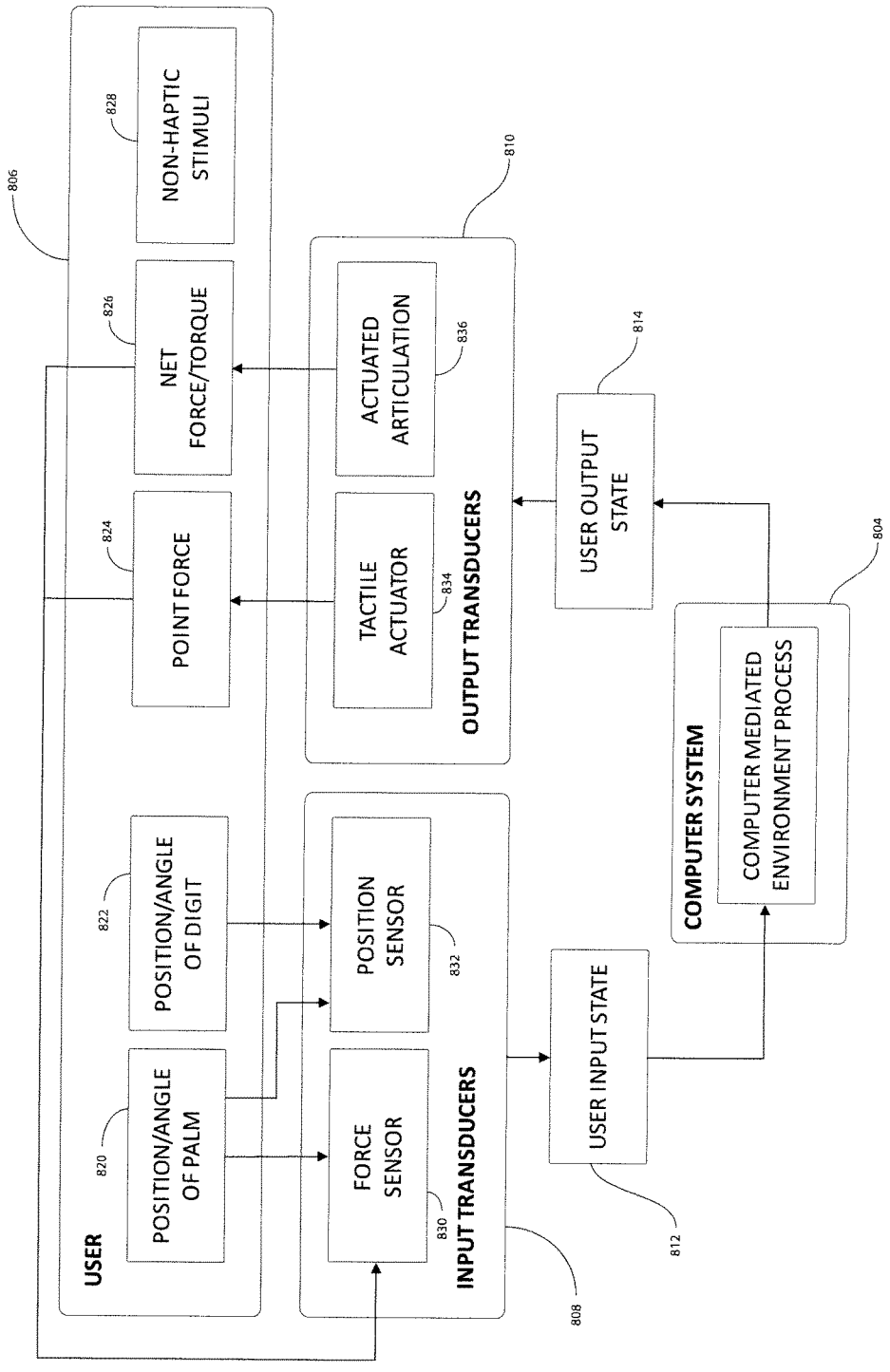

806
USER
820
POSITION/ANGLE OF PALM
822
POSITION/ANGLE OF DIGIT
824
POINT FORCE
826
NET FORCE/TORQUE
828
NON-HAPTIC STIMULI
808
INPUT TRANSDUCERS
830
FORCE SENSOR
832
POSITION SENSOR
812
USER INPUT STATE
810
OUTPUT TRANSDUCERS
834
TACTILE ACTUATOR
836
ACTUATED ARTICULATION
814
USER OUTPUT STATE
804
COMPUTER SYSTEM
COMPUTER MEDIATED ENVIRONMENT PROCESS

FIG. 8

HAPTIC FEEDBACK GLOVE

[0001] This application relates to U.S. patent application Ser. No. 15/372,362 entitled WHOLE-BODY HUMAN-COMPUTER INTERFACE, filed Dec. 7, 2016, which is a continuation of U.S. patent application Ser. No. 14/981,414, filed Dec. 28, 2015, which is a continuation of International Application No. PCT/US14/44735, filed Jun. 27, 2014, which claims the benefit of Provisional Application No. 61/843,317, filed Jul. 5, 2013, all of which are incorporated in their entirety herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates generally to human-machine interfaces to the hand, and more specifically to virtual reality human-machine interfaces to the hand. Even more specifically, the present invention relates to virtual reality human-machine interfaces to the hand that include cutaneous and kinesthetic feedback.

2. Discussion of the Related Art

[0003] Design of immersive virtual reality human-machine interfaces to the hand is a long-standing challenge. The dexterity, sensitivity, and small size of the human hand make it extremely difficult to design a virtual reality human-computer interface that permits natural hand interaction with computer-mediated environments.

[0004] U.S. patent application Ser. No. 15/372,362 describes a whole-body human-computer interface capable of simulating highly realistic interaction with virtual reality environments. The present invention comprises a series of improvements to the hand portion of the human-computer interface garment disclosed therein. Said hand portion will hereafter be referred to as a "haptic feedback glove."

[0005] Haptic feedback gloves have broad commercial applications, including in entertainment; medical and industrial training; and computer-aided design and manufacturing. Said applications broadly require haptic feedback gloves with the following combination of features absent in the present art:

[0006] Generality: human-machine interfaces to the hand of the present art, including haptic feedback gloves, are typically built and programmed for a single narrow range of applications. These systems employ simplified simulation parameters to achieve a design that is conducive to their particular application, but severely limited in general applicability. Such a design methodology reduces mechanical and computational complexity for some tasks, but at the cost of compromising flexibility, adaptability, and economy of scale of the resultant systems.

[0007] Realism: touch sensation is comprised of multiple sensory modalities described in the art. In particular, cutaneous feedback (mechanical stimulation of the skin), and kinesthetic feedback (net forces applied to the musculoskeletal system) are both critical for realistic touch sensation and natural interaction. Haptic feedback gloves of the present art typically only include a single sensory modality. Said devices of the present art also typically lack the resolution, displacement, frequency response, force output, or other performance characteristics required to realistically stimulate a particular sensory modality.

[0008] Practicality: to be commercially useful, haptic feedback gloves must be light and low-profile enough to be comfortably worn on the hand, and robust enough to survive repeated use in a real-world environment. They must also be low-cost enough to be commercially practical. Lastly, they must be able to be donned and doffed relatively quickly by a user. Haptic feedback gloves of the present art lack some or all of these qualities. Even the best performing devices of the known art (and in fact particularly the best performing devices) are simply impractical, as well as being substantially uneconomical. Even if these devices did overcome all of the shortcomings listed above, they would still likely be incapable of broad application due to their prohibitive cost and complexity.

[0009] Thus, there remains a significant need for an improved haptic feedback glove.

SUMMARY OF THE INVENTION

[0010] In accordance with one embodiment, the present invention can be characterized as a human-computer interface system including: a sensor configured to transduce the location of a finger of a hand of a user; an exoskeleton configured to exchange mechanical energy with the finger; an interface garment, including: an interface laminate coupled to a counterpressure assembly, configured to stimulate the user by applying a pressure to the finger; a first plurality of tactile actuators configured to apply a pressure to a finger of the user; and a second plurality of tactile actuators configured to apply a pressure to a palm of the hand of the user.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The above and other aspects, features and advantages of several embodiments of the present invention will be more apparent from the following more particular description thereof, presented in conjunction with the following drawings.

[0012] FIG. **1**A is a top view of a haptic feedback glove in accordance with one embodiment of the present invention.

[0013] FIG. **1**B is a bottom view of the haptic feedback glove of the embodiment of FIG. **1**A.

[0014] FIG. **2** is an exploded perspective view of a fingertip assembly of the haptic feedback glove of the embodiment of FIG. **1**A.

[0015] FIG. **3** is an exploded perspective view of an actuator interacting with a force transmission element of the haptic feedback glove of the embodiment of FIG. **1**A.

[0016] FIG. **4** is an exploded view of a hypothenar assembly of a haptic feedback glove in accordance with one embodiment.

[0017] FIG. **5** is a partial bottom view of a thenar assembly of a haptic feedback glove, in accordance with one embodiment, showing an interface laminate, and an armature and tensile members of a counterpressure assembly.

[0018] FIG. **6** is a partial bottom view of a hypothenar assembly of a haptic feedback glove, in accordance with one embodiment, showing an interface laminate, and an armature and tensile members of a counterpressure assembly.

[0019] FIG. **7** is a partial bottom view of an interdigital assembly of a haptic feedback glove, in accordance with one embodiment, showing an interface laminate, and an armature and tensile members of a counterpressure assembly.

[0020] FIG. **8** is a block diagram of a haptic feedback glove in accordance with one embodiment.

[0021] Certain components in the figures that are substantially identical across each finger (e.g. **135**, **133**, **202**, **261**, **252**, **250**, **254**, **258**, **260**, **262**, **206**, **132**, **207**, **204**, **205**, **208**, **118**, **119**, **120**, **121**, **122**, **123**, **124**, **125**, **146**, **142**, **143**, **140**, **148**, **315**, **316**, **314**, **202**, **320**, **308**, **310**, **324**, **326**, **306**, **304**, **207**) are given only a single label for clarity. References through the Detailed Description to these components should be understood to apply to said components of any or all fingers.

DETAILED DESCRIPTION

[0022] The following description is not to be taken in a limiting sense, but is made merely for the purpose of describing the general principles of exemplary embodiments. The scope of the invention should be determined with reference to the claims.

[0023] Reference throughout this specification to "one embodiment, "an embodiment," or similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, appearances of the phrases "in one embodiment," "in an embodiment," and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment.

[0024] Furthermore, the described features, structures, or characteristics of the invention may be combined in any suitable manner in one or more embodiments. In the following description, numerous specific details are presented to provide a thorough understanding of embodiments of the invention. One skilled in the relevant art will recognize that the invention can be practiced without one or more of the specific details, or with other methods, components, materials, and so forth. In other instances, well-known structures, materials, or operations are not shown or not described in detail to avoid obscuring aspects of the invention.

Definitions and Conventions

[0025] Definitions and conventions are identical to those in U.S. patent application Ser. No. 15/372,362, unless otherwise specified.

[0026] As used herein, the term "haptic feedback glove" means: a hand portion of a human-computer interface garment.

[0027] As used herein, the term "finger" means: a digit of the hand, including the thumb. "Digit" and "finger" are used interchangeably throughout the present application.

[0028] As used herein, the term "mechanical ground" means: a point that is substantially fixed and immovable with respect to a finger of the user, rather than with respect to the user as a whole as defined in U.S. patent application Ser. No. 15/372,362.

[0029] As used herein, the term "position sensor" means: a sensor configured to detect at least one of position and orientation.

[0030] As used herein, the term "force sensor" means: a sensor configured to detect at least one of force and torque.

Overview

[0031] FIG. **8** shows a block diagram of a haptic feedback glove in accordance with one embodiment of the present invention. Shown is a plurality of input transducers **808** and output transducers **810** coupled to a computer system **804** and to a user **806**. The input transducers **808** receive input from the user **806**, and transduce that input to a user input state **812** preferably defined at a discrete time step n. The output transducers **810** receive a user output state **814** from the computer system **804**, preferably defined at a discrete time step n+1. The user output state **814** is transduced by the output transducers **810** to an appropriate form so as to stimulate one or more of the user's **806** sensory systems. Non-haptic stimuli **828** (e.g. visual, auditory, or chemosensory stimuli) are preferably synchronized with haptic stimuli provided by the haptic feedback glove as described in U.S. patent application Ser. No. 15/372,362.

[0032] FIGS. **1**A and **1**B show a top and bottom view respectively of a haptic feedback glove in accordance with one embodiment of the present invention. The haptic feedback glove comprises an interface garment, including an interface laminate and an exoskeleton. Position sensors transduce the position of the user's digits. Preferably, an additional position sensor transduces the position of the user's palm.

Interface Laminate

[0033] In a preferred embodiment, a haptic feedback glove comprises an interface laminate comprising a plurality of tactile actuators **834** (FIG. **8**) coupled to the skin of the user's hand. A counterpressure assembly provides a normal force counter to the force produced by said interface laminate against the user's skin, holding the interface laminate against the skin during actuation of tactile actuators of the laminate and motion of the user's hand.

Fingertip Assembly

[0034] Referring to FIG. **2**, a fingertip assembly **200** of a haptic feedback glove preferably comprises an interface laminate segment **204** located such that tactile actuators **205** are coupled to substantially all of the user's finger pad. In a preferred embodiment, fingertip assembly **200** comprises at least 12 tactile actuators **205** contacting said finger pad. In a more preferred embodiment, fingertip assembly **200** comprises at least 24 tactile actuators contacting said finger pad. In a preferred embodiment, tactile actuators **205** of fingertip assembly **200** are configured to produce a displacement of at least 0.5 mm. In a more preferred embodiment, tactile actuators **205** of fingertip assembly **200** are configured to produce a displacement of at least 1 mm.

[0035] The interior surface of interface laminate segment **204** is coupled to intermediate layer **208**. Intermediate layer **208** is in turn coupled to the user's hand. The exterior surface of interface laminate segment **204** is coupled to the interior surface of fingertip counterpressure assembly **250**. Ribbon assembly **207** of interface laminate segment **204** preferably exits the proximal side of fingertip counterpressure assembly **250**, above the user's nail.

[0036] Fingertip counterpressure assembly **250** comprises an armature **256** and a tensile member **258**. Armature **256** provides structural support to fingertip assembly **200**, helping interface laminate segment **204** remain in continuous contact with the user's finger. Armature **256** is coupled to interface laminate segment **204** by means of a suitable adhesive, preferably room-temperature vulcanization silicone. In one embodiment, armature **256** comprises an elastomer, such as polydimethylsiloxane or fiberglass-reinforced

silicone. In another embodiment, armature **256** comprises a non-elastomeric polymer, such as high-density polyethylene or polyimide.

[0037] Armature **256** is coupled to tensile member **258**, which resists forces applied orthogonal to the interior surface of armature **256** by interface laminate segment **204**. Tensile member **258** preferably comprises an elastic textile, such as Lycra, an elastomer, such as latex, or another suitable high strain material. Tensile member **258** serves a secondary purpose as a donning aid, enabling fingertip assembly **200** to stretch to accommodate various finger sizes, while retaining a suitable level of counterpressure for the operation of interface laminate segment **204**. Sensor mounting point **252** is a declivity in the upper surface of armature **256** that provides a mounting point for magnetic position sensor **135**, and aids in the routing of position sensor leads **133**.

[0038] Armature **256** is preferably shaped such that it closely matches the curvature of the user's fingertip. In one embodiment, fingertip armature **256** is slightly undersized relative to the distal phalange of the target user, such that it produces a nominal force against the top and bottom of the finger when worn to help interface laminate segment **204** remain in contact with the fingertip. In a preferred embodiment, Armature **256** does not extend to the medial and lateral sides of the finger to avoid interfering with finger ad- and abduction. Similarly, the bottom proximal portion of armature **256** ends distally enough to the distal interphalangeal crease to avoid interfering with the motion of the distal interphalangeal joint.

[0039] Embodiments are contemplated in which tactile interface laminate segments are also coupled to the pads of the intermediate and proximal phalanges of one or more of the user's fingers, in addition to the pad of the distal phalange. Embodiments are also contemplated in which tactile interface laminate segments extend to the dorsal, medial, or lateral aspects of the fingers. One or more separate segments may also be employed to extend sensation to the dorsal, medial, or lateral aspects of the fingers, rather than extending the existing segments.

[0040] Fingertip assembly **200** is coupled to fabric substrate **115** to facilitate ease of donning. Fabric substrate **115** preferably comprises Lycra or another lightweight, elastic fabric. While alternate embodiments are contemplated in which fingertip assemblies are donned as separate elements, in the preferred embodiment, fingertip assemblies are all coupled to fabric substrate **115** such that they can be donned with a single motion, like a typical glove.

[0041] Interface laminate segment **204** is coupled to a vibration actuator **206**. In a preferred embodiment, vibration actuator **206** is configured to produce vibrations of 20 Hz-300 Hz that are detectible by the user through interface laminate segment **204** and intermediate layer **208**. In a more preferred embodiment, vibration actuator **206** is configured to produce vibrations of 20 Hz-1 kHz that are detectible by the user through interface laminate segment **204**.

[0042] Vibration actuator **206** preferably comprises a multi-layer piezoelectric actuator, such as a piezoceramic or piezopolymer actuator, a non-piezoelectric electroactive polymer actuator, or another suitable solid state actuator. In a first alternate embodiment, vibration actuator **206** comprises an electromechanical actuator, such as an eccentric rotating mass, linear resonant actuator, or other vibration motor. In a second alternate embodiment, vibration actuator **206** comprises a fluidic actuator.

[0043] A portion of vibration actuator **206** is coupled to inner surface **262** of fingertip armature **256**, such that a majority of vibration actuator **206** is still permitted free motion relative to inner surface **262**. An air gap is preferably left between the inner surface of vibration actuator **206** and the outer surface of interface laminate segment **204**. In an alternate embodiment, vibration actuator **206** is placed on the top, rather than the bottom, portion of fingertip armature **256**, such that it contacts the user's fingernail. In this embodiment, vibrations are transmitted through the fingernail and distal phalange into the finger pulp.

[0044] Vibration actuator **206** is coupled to vibration actuator leads **132**, which supply electric power to vibration actuator **206**. Vibration actuator leads **132** preferably run along the back of the user's finger, as shown in FIG. **1**A, in accordance with one embodiment.

Palm Assembly

[0045] Referring to FIG. **1**B, a palm portion of a haptic feedback glove comprises a plurality of palm assemblies—thenar assembly **150**, hypothenar assembly **160**, and interdigital assembly **180**—configured to permit uninhibited movement of the user's hand while remaining in contact with as much of the user's palm as possible. Thenar assembly **150** contacts the thenar eminence of the user's palm. Hypothenar assembly **160** contacts the hypothenar eminence of the user's palm. Interdigital assembly **180** sits between the transverse creases (distal and proximal) of the user's palm and the palmar interdigital creases of the user's fingers, in the interdigital region of the palm. The thenar crease and distal and proximal transverse creases are left deliberately free of material to maximize hand mobility. Alternate embodiments of a haptic feedback glove are contemplated wherein more or less than three palm assemblies are employed.

Hypothenar Assembly

[0046] FIG. **4** shows an exploded view of hypothenar assembly **160**, in accordance with one embodiment. Hypothenar assembly **160** comprises interface laminate segment **163**, coupled to the skin of the user's hypothenar eminence by means of intermediate layer **161**. In a preferred embodiment, interface laminate segment **163** comprises a tactile actuator density of at least 0.75 actuators per square centimeter. In a more preferred embodiment, interface laminate segment **163** comprises a tactile actuator density of at least 1.50 actuators per square centimeter. In a preferred embodiment, tactile actuators of interface laminate segment **163** are configured to produce a displacement of at least 1 mm. In a more preferred embodiment, tactile actuators of interface laminate segment **163** are configured to produce a displacement of at least 2 mm.

[0047] The bottom surface of interface laminate segment **163** is coupled to intermediate layer **161**. Intermediate layer **161** is in turn coupled to the user's hand. The top surface of interface laminate segment **163** is coupled to the bottom surface of hypothenar counterpressure assembly **175**. Ribbon assembly **179** of interface laminate segment **163** preferably exits the proximal side of hypothenar counterpressure assembly **175**, being routed through wrist strap **116**.

[0048] Hypothenar counterpressure assembly **175** comprises an armature **165** and a plurality of tensile members **162**, **164**, **166**. Armature **165** provides structural support to hypothenar assembly **160**, helping interface laminate segment **163** remain in continuous contact with the user's hypothenar eminence. Armature **165** is coupled to interface laminate segment **163** by means of a suitable adhesive, preferably room-temperature vulcanization silicone **169**. In one embodiment, armature **165** comprises an elastomer, such as polydimethylsiloxane or fiberglass-reinforced silicone. In another embodiment, armature **165** comprises a non-elastomeric polymer, such as high-density polyethylene or polyimide.

[0049] FIG. **6** shows the geometry of hypothenar assembly **160** in a top and side view respectively, in accordance with one embodiment. Hypothenar assembly **160** is generally shaped to match the shape and curvature of the user's hypothenar eminence. Distolateral tip **604** of hypothenar assembly **160** is preferably flared slightly above the surface of the palm to aid in the generation of a counterforce against the portion of the user's palm under said tip.

[0050] Referring now to FIGS. **1**A and **1**B, armature **165** is coupled to tensile members **162**, **164**, **166** which resist forces applied orthogonal to the interior surface of armature **165** by interface laminate segment **163**. Tensile members **162**, **164**, **166** preferably comprise an elastic textile, such as Lycra. Said tensile members **162**, **164**, **166** can also comprise an elastomer, such as latex, or another suitable high strain material. Tensile members **162**, **164**, **166** serve a secondary purpose as a donning aid, enabling hypothenar assembly **160** to stretch to accommodate various hand sizes, while retaining a suitable level of counterpressure for the operation of interface laminate segment **163**.

[0051] Tensile member **162** is routed through a cutout **174** in fabric substrate **115**, over the top of the user's thenar eminence, through the thenar space, and is coupled to opisthenar plate **111** located on the back of the user's hand. Opisthenar plate **111** comprises any suitable rigid structural material, preferably a polymer or fiber-reinforced polymer composite. Tensile member **166** is routed through a cutout **178** in fabric substrate **115**, medially over the blade of the palm to couple to the medial side of opisthenar plate **111**. Tensile member **164** is routed proximally through a cutout **176** in fabric substrate **115** and coupled to wrist strap **116**, which substantially encircles the user's wrist.

[0052] Interface laminate segment **163** is coupled to vibration actuator assemblies **170**, **172**. Vibration actuator assemblies **170**, **172** comprise a casing and a vibration actuator. In a preferred embodiment, said vibration actuators are configured to produce vibrations of 20 Hz-300 Hz that are detectible by the user through interface laminate segment **163** and intermediate layer **161**. In a more preferred embodiment, said vibration actuators are configured to produce vibrations of 20 Hz-1 kHz that are detectible by the user through interface laminate segment **163** and intermediate layer **161**.

[0053] Vibration actuators of vibration actuator assemblies **170**, **172** preferably comprise a multi-layer piezoelectric actuator, such as a piezoceramic or piezopolymer actuator, a non-piezoelectric electroactive polymer actuator, or another suitable solid state actuator. In an alternate embodiment, said vibration actuators comprise an electromechanical actuator, such as an eccentric rotating mass, linear resonant actuator, or other vibration motor.

[0054] A portion of the vibration actuators of vibration actuator assemblies **170**, **172** is coupled to the upper inner surface of their casings, such that a majority of said vibration actuators are still permitted free motion relative to said upper inner surfaces. An air gap is preferably left between the bottom surface of the vibration actuators and the top surface of interface laminate segment **163**.

[0055] Vibration actuator assemblies **170**, **172** are preferably spaced substantially evenly across the surface area of the user's hypothenar eminence, with a density of least 0.1 actuators per square centimeter. In a more preferred embodiment, vibration actuators have a density of least 0.2 actuators per square centimeter. Vibration actuators of vibration actuator assemblies **170**, **172** are coupled to vibration actuator leads **171**, **173** which supply electric power to the vibration actuators of said assemblies. Vibration actuator leads **171**, **173** preferably run along the user's palm to the wrist.

[0056] Embodiments are contemplated in which tactile interface laminate segments extend to the dorsal or medial aspects of the hypothenar eminence. One or more separate segments may also be employed to extend sensation to dorsal or medial aspects of the hypothenar eminence, rather than extending the existing segments. As with fingertip assembly **200**, hypothenar assembly **160** is coupled to fabric substrate **115** to facilitate ease of donning.

Thenar Assembly

[0057] Referring now to FIGS. **1**B and **5**, thenar assembly **150** comprises interface laminate segment **153**, coupled to the skin of the user's thenar eminence by means of an intermediate layer (not shown). Tactile actuator density and displacement of interface laminate segment **153** are similar to interface laminate segment **163** of hypothenar assembly **160**. Tactile actuator displacement and density can vary between palm assemblies. For instance, the tactile actuator density of interface laminate segment **153** can be slightly higher than that of interface laminate segment **163** due to increased tactile sensitivity in the thenar region.

[0058] The bottom surface of interface laminate segment **153** is coupled to an intermediate layer (not shown), which is in turn coupled to the user's hand. The top surface of interface laminate segment **153** is coupled to the bottom surface of thenar counterpressure assembly **158**. Ribbon assembly **159** of interface laminate segment **153** preferably exits the proximal side of thenar counterpressure assembly **158**, being routed proximally through wrist strap **116**.

[0059] As with hypothenar counterpressure assembly **175**, thenar counterpressure assembly **158** comprises an armature **155** and a plurality of tensile members **152**, **154**, **156** of substantially identical composition to those of hypothenar counterpressure assembly **175**. FIG. **5** shows the geometry of thenar assembly **150** in a top and projected view respectively, in accordance with one embodiment. Thenar assembly **150** is generally shaped to match the shape and curvature of the user's thenar eminence. Radius of curvature **502** is preferably slightly smaller than the corresponding radius of curvature of the user's thenar eminence. Said difference in radius of curvature provides a nominal force against the user's thenar eminence when thenar assembly **150** is worn to help interface laminate segment **153** remain in contact with the thenar eminence.

[0060] In a preferred embodiment, the stiffness of thenar assembly **150** varies across its surface. Said variation in

stiffness minimizes interference with thenar motion, particularly around the thenar crease and palmar interdigital crease of the thumb, while maintaining sufficient structural integrity to provide effective counterpressure for interface laminate segment **153**. The portion of thenar assembly **150** bonded to armature **155** has a higher stiffness than remaining portions comprising only the interface laminate. Armature **155** does not extend all the way to the thenar crease and palmar interdigital crease of the thumb, creating a more compliant region in thenar assembly **150** around these highly mobile areas.

[0061] Referring now to FIGS. **1**A and **1**B, armature **155** is coupled to tensile members **152**, **154**, **156** which resist forces applied orthogonal to the interior surface of armature **155** by interface laminate segment **153**. Tensile members **152**, **154**, **156** are of substantially identical composition and purpose to those of hypothenar assembly **160**.

[0062] Tensile member **152** is routed over the top of the user's thenar eminence, through the thenar space, and is coupled to thenar plate **114** located on the dorsum of the user's thumb. Thenar plate **114** comprises any suitable rigid structural material, preferably a polymer or fiber-reinforced polymer composite. Tensile member **154** is coupled to thenar plate **114** above the first metacarpal. Thenar plate **114** is preferably coupled to opisthenar plate **111** by means of tensile member **157**. Tensile member **156** is routed proximally through wrist strap **116**.

[0063] Thenar assembly **150** comprises vibration actuators substantially identical to those of hypothenar assembly **160**. Said thenar vibration actuators are preferably spaced substantially evenly across the surface area of the user's thenar eminence, with a density similar to the vibration actuators of the hypothenar assembly **160**.

[0064] Embodiments are contemplated in which tactile interface laminate segments extend to the dorsal, lateral, or proximal aspects of the thenar eminence. One or more separate segments may also be employed to extend sensation to dorsal, lateral, or proximal aspects of the thenar eminence, rather than extending the existing segments. As with fingertip assembly **200**, and hypothenar assembly **160**, thenar assembly **150** is coupled to fabric substrate **115** to facilitate ease of donning.

Interdigital Assembly

[0065] Referring now to FIGS. **1**B and **7**, interdigital assembly **180** comprises interface laminate segment **183**, coupled to the skin of the interdigital region of the user's palm by means of an intermediate layer (not shown). Tactile actuator density and displacement of interface laminate segment **183** are similar to interface laminate segment **163** of hypothenar assembly **160**. Tactile actuator displacement and density can vary between palm assemblies. For instance, the tactile actuator displacement of interface laminate segment **183** can be slightly lower than that of interface laminate segment **163**.

[0066] The bottom surface of interface laminate segment **183** is coupled to an intermediate layer (not shown), which is in turn coupled to the user's hand. The top surface of interface laminate segment **183** is coupled to the bottom surface of interdigital counterpressure assembly **188**. Ribbon assembly **189** of interface laminate segment **183** preferably exits the medial side of interdigital counterpressure assembly **188**, being routed around the fifth metacarpal to the dorsal aspect of the hand, then along the back of the hand to the wrist.

[0067] As with hypothenar counterpressure assembly **175**, and thenar counterpressure assembly **158**, interdigital counterpressure assembly **188** comprises an armature **185** and a plurality of tensile members **182**, **184** of substantially identical composition to said other palm assemblies. FIG. **7** shows the geometry of interdigital assembly **180** in a top and projected view respectively, in accordance with one embodiment. Interdigital assembly **180** is generally shaped to match the shape and curvature of the interdigital region of the user's palm. Radius of curvature **702** is preferably slightly smaller than the corresponding radius of curvature of the user's palm. Said difference in radius of curvature provides a nominal force against the user's palm when interdigital assembly **180** is worn to help interface laminate segment **183** remain in contact with the user's palm.

[0068] In a preferred embodiment, the stiffness of interdigital assembly **180** varies across its surface. Said variation in stiffness minimizes interference with the motion of the index, middle, ring, and pinky fingers, particularly around the distal and proximal transverse creases and the palmar interdigital creases of said fingers, while maintaining sufficient structural integrity to provide effective counterpressure for interface laminate segment **183**. The portion of interdigital assembly **180** bonded to armature **185** has a higher stiffness than remaining portions comprising only the interface laminate. Armature **185** does not extend all the way to the distal and proximal transverse creases and the palmar interdigital creases of the index, middle, ring, and pinky fingers, creating a more compliant region in interdigital assembly **180** around these highly mobile areas.

[0069] Referring now to FIGS. **1**A and **1**B, armature **185** is coupled to tensile members **182**, **184** which resist forces applied orthogonal to the interior surface of armature **185** by interface laminate segment **183**. Tensile members **182**, **184** are of substantially identical composition and purpose to those of thenar assembly **150** and hypothenar assembly **160**.

[0070] Tensile member **184** is routed medially over the fifth metacarpal to couple to the medial side of opisthenar plate **111** located on the back of the user's hand. Tensile member **182** is routed laterally over the second metacarpal to couple to the lateral side of opisthenar plate **111**.

[0071] Interdigital assembly **180** comprises vibration actuators substantially identical to those of thenar assembly **150**, and hypothenar assembly **160**. Said interdigital vibration actuators are preferably spaced substantially evenly across the surface area of the interdigital region of the user's palm, with a density similar to the vibration actuators of the thenar and hypothenar assemblies.

[0072] Embodiments are contemplated in which tactile interface laminate segments extend to the dorsal, medial, or lateral aspects of the interdigital region of the user's palm. One or more separate segments may also be employed to extend sensation to dorsal, medial, or lateral aspects of the interdigital region, rather than extending the existing segments. As with fingertip assembly **200**, thenar assembly **150**, and hypothenar assembly **160**, interdigital assembly **180** is coupled to fabric substrate **115** to facilitate ease of donning.

Exoskeleton

[0073] In a preferred embodiment, a haptic feedback glove comprises an exoskeleton, said exoskeleton comprising a

plurality of actuated articulations **836** (FIG. **8**). Referring now to FIG. **1**A, an exoskeleton of a haptic feedback glove is shown, in accordance with one embodiment. Said exoskeleton comprises: a finger exoskeleton assembly **107**, including an actuator assembly **300**, a kinematic termination **190**, a force transmission element **202**, and a mechanical ground connection **304**.

[0074] Force transmission element **202** is mechanically coupled to kinematic termination **190**, and is variably coupled to mechanical ground connection **304** by means of actuator **308** (FIG. **3**). Enabling actuator **308** (FIG. **3**) modifies the net force on the user's finger, preferably by means of modifying the physically-defined impedance of finger exoskeleton assembly **107** by controlling the extent of mechanical coupling between force transmission element **202** and mechanical ground connection **304**.

Kinematic Termination

[0075] FIG. **2** shows an exploded perspective view of the fingertip assembly **200** of a haptic feedback glove, in accordance with one embodiment, comprising a kinematic termination **190**. Kinematic termination **190** comprises a load path from a user's fingertip to force transmission element **202**, comprising: intermediate layer **208**; coupled to interface laminate segment **204**; coupled in turn to fingertip counterpressure assembly **250**; and finally, to force transmission element **202**, by means of projection **260**, through-hole **261**, and guide slot **254**.

[0076] The compliance of fingertip counterpressure assembly **250** can be tuned to optimize the balance between stiffness of kinematic termination **190** and the ability to accommodate a wide range of finger sizes. A more compliant fingertip counterpressure assembly **250** will enable fingertip assembly **200** to stretch to fit a greater range of finger sizes, while still providing effective counterpressure, at the expense of reducing the effective stiffness of the constraint to finger motion imposed by finger exoskeleton assembly **107** (FIG. **1**A).

[0077] Finger motion acting against the finger exoskeleton assembly **107** results in reaction forces which are distributed via the load path of kinematic termination **190** to the user's finger. Preferably, this termination of reaction forces occurs at the distal phalange of the finger. More preferably, this termination of reaction forces is distributed approximately evenly across the palmar surface of said phalange. The kinematic termination **190** is preferably shaped to minimize interference with finger motion and interference with a wearer's workspace, particularly.

[0078] In a preferred embodiment, interface laminate segment **204** acts to distribute the net force on the user's fingertip produced by the action of finger exoskeleton assembly **107** via kinematic termination **190**, such that point forces at the fingertip approximate the physical point forces resulting from a particular object interaction. For example, pressing on a simulated pin and a simulated flat surface in a virtual environment might produce identical net forces on the user's fingertip, as rendered by the action of finger exoskeleton assembly **107**; however, these interactions would produce very different point forces on the skin of the fingertip as rendered by the action of tactile actuators **205** of interface laminate segment **204**.

Force Transmission Element

[0079] Referring to FIG. **1**A, the action of finger exoskeleton assembly **107** results in forces which are transmitted from kinematic termination **190** to mechanical ground connection **304** via force transmission element **202**. Force transmission element **202** can be designed to transmit both tensile and compressive forces (e.g. a continuous mechanical linkage), compressive forces only (e.g. a series of disconnected linkages sharing a common centerline), or tensile forces only (e.g. a cable).

[0080] In a preferred embodiment shown in FIG. **1**A, force transmission element **202** is a tendon located on the dorsum of the user's hand that transmits tensile forces, applying forces to the user's finger during grasping motions involving finger flexion while allowing unhindered finger extension. Force transmission element **202** is preferably ribbon shaped (i.e. having a ratio of width to thickness of at least 10), and composed of nylon or another suitable polymer or non-polymer material with minimal elongation under tensile load, a smooth surface finish, flexibility under bending load, high toughness, and a low coefficient of friction relative to any bearing surfaces—e.g. actuator casing lower lip **307**, and upper lip **315** (FIG. **3**), or force transmission element guide slots **119**, **121**.

[0081] Numerous alternate cross-sectional geometries of a force transmission element are contemplated, including circular, elliptical, and multi-body or multi-stranded cross sections. Cross section can vary across the length of a force transmission element. For example, the portion of force transmission element **202** contacting actuator assembly **300** can be ribbon shaped to maximize contact area between the actuator assembly **300** and the force transmission element **202**, while other portions of the force transmission element **202** have a circular cross section.

[0082] Force transmission element **202** is coupled to the user's finger by means of kinematic termination **190**. Said force transmission element **202** is then coupled to force transmission guides **118** and **120**, which are in turn coupled to the intermediate and proximal phalanges of the user's finger. Force transmission element **202** is free to slide proximodistally via force transmission element guide slots **119**, **121** in force transmission guides **118**, **120**, but is substantially fully constrained in all other axes of motion. During flexion of the user's finger, when finger exoskeleton assembly **107** is active, force transmission element **202** applies a compressive load to structural members **123**, **125**, which in turn apply an equal compressive load to the top of the user's finger. The height of force transmission element guide slots **119**, **121** relative to the user's finger strongly influences both the magnitude and vector of force that will be applied to the kinematic termination **190** for a given tensile force on force transmission element **202** and a given position of the user's finger. In a preferred embodiment, said height of force transmission element guide slots **119**, **121** is greater than 0.5 cm and less than 5 cm. In a more preferred embodiment, said height of force transmission element guide slots **119**, **121** is greater than 1 cm and less than 2.5 cm.

[0083] Elastic straps **122**, **124** secure force transmission guides **118**, **120** to the intermediate and proximal phalanges of the user's finger. Said elastic straps **122**, **124** are preferably composed of an elastic fabric, such as Lycra, but can also be composed of an elastomer or other suitable elastic material. As with fingertip counterpressure assembly **250**, elastic straps **122**, **124** are preferably coupled to fabric substrate **115** to facilitate the donning of a haptic feedback glove as a single unit, in the manner of a typical glove.

[0084] In one embodiment, force transmission element **202** is coupled to a vibration actuator, of any of the types described above, located proximally to fingertip assembly **200**. Vibrations from said actuator are transmitted to fingertip assembly **200** by means of force transmission element **202**, particularly when under tension.

Actuator

[0085] Referring still to FIG. **1**A, force transmission element **202** is variably coupled to actuator assembly **300** by means of actuator **308**, and finally to tensioning mechanism **143**. FIG. **3** shows an exploded perspective view of an actuator **308** interacting with a force transmission element **202**, in accordance with one embodiment. An actuator assembly **300** comprises an actuator **308**. Actuator **308** is configured to produce a variable force or displacement. In the preferred embodiment of FIG. **3**, actuator **308** is a miniature fluidic actuator constructed in a similar manner to a fluidic actuator of a tactile actuation laminate (as detailed in U.S. patent application Ser. No. 15/372,362).

[0086] Actuator **308** comprises an elastic membrane **320** bonded to a substrate **322** to form a plurality of actuation chambers **310**, **324**, **326**. A pressurized fluid is supplied to actuator **308** by means of tube **131**, via supply orifice **328**. Elastic membrane **320** can be controllably actuated by regulating the volume or pressure of working fluid flowing into and out of actuation chambers **310**, **324**, **326**.

[0087] Alternate embodiments are contemplated in which actuator **308** comprises a solid-state actuator (such as a piezoceramic or piezopolymer, or non-piezoelectric electroactive polymer actuator), an electromechanical actuator (such as a solenoid, voice coil, a brushed or brushless DC motor, or an AC induction or synchronous motor), or any other suitable actuator detailed in in U.S. patent application Ser. No. 15/372,362. Actuator **308** is preferably configured to produce a force of at least 10 N. In a more preferred embodiment, actuator **308** is configured to produce a force of at least 25 N. Actuator **308** is preferably configured to produce a displacement of at least 0.5 mm. The upper surface of actuator **308** preferably comprises an elastic material with a high coefficient of friction in contact with the material of force transmission element **202**, such as polydimethylsiloxane or other silicone-based or non-silicone-based elastomers.

[0088] In the preferred embodiment, as illustrated in FIG. **3**, actuator **308** is configured as a variable mechanical impedance brake. Alternate embodiments are contemplated wherein actuator **308** comprises an active actuation element configured to apply a variable force to force transmission element **202**, in addition to said variable impedance brake (in mechanical serial or parallel configurations), or instead of it. This alternate embodiment and other suitable embodiments are described in detail in U.S. patent application Ser. No. 15/372,362.

[0089] Actuator housing **192** (FIG. **1**) comprises upper housing **316** coupled to lower housing **306**. Actuator **308** is coupled to the upper face of lower housing **306**. Traction membrane **314** is coupled to the lower face of upper housing **316**. In actuator's **308** off state, actuator housing **192** (FIG. **1**) is coupled to force transmission element **202** by means of lower lip **307** and upper lip **315**, thus preventing direct contact between force transmission element **202** and actuator **308** or traction membrane **314**. In actuator's **308** on state, force transmission element **202** is frictionally coupled to actuator **308** and traction membrane **314** by the orthogonal displacement of actuator **308**. Actuator **308** can be configured for binary control—acting like a simple brake—or proportional control—allowing the application of multiple intermediate levels of force to force transmission element **202** between full on and full off.

[0090] Actuator housing **192** (FIG. **1**) supports actuator **308** and traction membrane **314**, applying a normal force during actuation to maintain contact between actuator **308**, traction membrane **314**, and force transmission element **202**. Traction membrane **314** is preferably included in actuator assembly **300** to maximize the holding force between actuator **308** and force transmission element **202**. In one embodiment, traction membrane **314** comprises a material similar to the upper surface of actuator **308**. In another embodiment, traction membrane **314** comprises a ratchet-like mechanism, having teeth or other projections that mate with similar projections on the surface of force transmission element **202** to increase the effective coefficient of friction between the two surfaces. In an alternate embodiment, traction membrane **314** is replaced with a second actuator above force transmission element **202**.

[0091] Preferably, the opening in actuator housing **192** (FIG. **1**) formed by upper lip **315** and lower lip **307** has a width sufficient to allow for some angular play of force transmission element **202** as the user's finger ad- and abducts. Said angular play permits substantially unobstructed ad- and abduction of the user's finger in free space. Actuator housing **192** (FIG. **1**) is coupled to mechanical ground by means of mechanical ground connection **304**.

[0092] Referring now to FIG. **1**, a single finger exoskeleton assembly **107** is shown coupled to the user's thumb. In an alternate embodiment, a second finger exoskeleton assembly is additionally coupled to the thumb to control motion of the metacarpal.

Mechanical Ground Connection

[0093] Reaction forces from the action of actuator **308** must be transferred to mechanical ground to produce a net force on the user's finger. In the embodiment of FIG. **1**A, mechanical ground comprises the user's palm and wrist. Reaction forces from the action of actuator **308** are transferred to opisthenar plate **111** by means of mechanical ground connection **304**, and in turn into the user's palm and wrist by means of fabric substrate **115** and wrist strap **116**, among other elements. Said armature preferably distributes reaction forces generated by actuator **308** across the skin surface of the user's hand and arm as evenly as possible to minimize anomalous point forces.

[0094] In an alternate embodiment, reaction forces from the action of actuator **308** are transferred to an external mechanical structure, such as an arm exoskeleton, preferably by means of a temporary coupling point. Several related embodiments are presented in U.S. patent application Ser. No. 15/372,362.

Tensioning Mechanism

[0095] FIG. **1**A shows a preferred embodiment of a finger exoskeleton assembly **107**, comprising a tensioning mechanism **143**. Said tensioning mechanism is coupled to force transmission element **202**, and serves to keep it under a nominal amount of tension during the operation of finger exoskeleton assembly **107**. In the absence of a tensioning

element, there is a risk of slack developing in force transmission element **202** distal to actuator assembly **300**. Any such slack will produce an undesirable delay between the onset of actuation and the application of forces to the user's finger.

[0096] Tensioning mechanism **143** comprises an elastic band **142**, preferably composed of natural latex or another suitable elastomer, coupled to force transmission element **202** by means of through-hole **146**. The proximal end of elastic band **142** is coupled to hook **148**, which is in turn coupled to termination block **140**. Termination block **140** is coupled to opisthenar plate **111**, and in turn to the user's hand by means of fabric substrate **115** and wrist strap **116**, among other elements.

[0097] In a preferred embodiment, tensioning mechanism **143**, or a separate tensioning mechanism, are additionally coupled to interface laminate ribbon assembly **207** to minimize slack in said ribbon assembly **207**. Unlike force transmission element **202**, slack in interface laminate ribbon assembly **207** won't compromise the function of interface laminate segment **204**; however, slack in ribbon assembly **207** is undesirable as it will cause ribbon assembly **207** to protrude unnecessarily above the user's fingers during flexion. This protrusion increases the profile of a haptic feedback glove and introduces a risk of the user's fingers becoming tangled in ribbon assembly **207**.

[0098] In an alternate embodiment, the tensioning mechanism **143** comprises an actuator of any of the types contemplated above or in U.S. patent application Ser. No. 15/372,362, which actively controls the tension of force transmission element **202**. Said actuator can also be used to actuate finger exoskeleton assembly **107**, as described above.

Sensors

[0099] FIG. **8** shows a block diagram of a haptic feedback glove, in accordance with one embodiment. A haptic feedback glove preferably comprises a position sensor **832** configured to transduce the position or angle **822** of a digit of the user's hand, and a position sensor **832** configured to transduce the position or angle **820** of the palm of the user's hand. Many position sensors **832** suitable for capturing hand motion are described in U.S. patent application Ser. No. 15/372,362.

[0100] A haptic feedback glove can also comprise one or more force sensors **830** configured to transduce a point force **824** on the user's skin, or a net force/torque **826** on a digit of the user's hand to enable closed loop force control. Many suitable force sensors **830** are described in U.S. patent application Ser. No. 15/372,362.

[0101] In a preferred embodiment, shown in FIG. **1**A, a magnetic position sensor **135** on the fingertip, and a magnetic position sensor (not shown) on the palm are used to transduce hand position. In an alternate embodiment, the magnetic sensor on the palm is replaced with an optical sensor. In a second alternate embodiment, wherein the haptic feedback glove is coupled to an arm exoskeleton, the magnetic sensor on the palm is obviated by position information supplied by the arm exoskeleton. In one variation, a first palm sensor is placed between the second and third metacarpals, and a second palm sensor is placed between the fourth and fifth metacarpals to transduce motion of the carpometacarpal joints of the pinky and/or ring finger.

[0102] In a preferred embodiment, a magnetic emitter (not shown) coupled to the user's hand emits a magnetic field. The strength of the magnetic field is employed by magnetic position sensor **135** to transduce its position relative to the emitter. Such magnetic emitters are well known in the art. In an alternate embodiment, said emitter is located off the user's body.

[0103] In one embodiment, biosignal sensors are included in the haptic feedback glove, as described in U.S. patent application Ser. No. 15/372,362.

Veneer and Undersuit

[0104] In a preferred embodiment, a veneer layer (not shown) is included over a haptic feedback glove. In one embodiment, this veneer layer simply comprises a thin fabric glove. In another embodiment, the veneer layer includes rigid elements, particularly on the dorsal surface of the hand. The veneer layer serves to protect the functional components of a haptic feedback glove during operation and enhance the aesthetic appeal of the haptic feedback glove.

[0105] In a preferred embodiment, an undersuit glove is donned by the user before donning a haptic feedback glove. Said undersuit glove prevents direct skin contact between the user and the inside of the haptic feedback glove. The use of an undersuit glove reduces the need to clean the haptic feedback glove, and offers improved hygiene, particularly in cases where a single haptic feedback glove is shared between multiple users. Said undersuit is described in greater detail in U.S. patent application Ser. No. 15/372,362.

[0106] While the invention herein disclosed has been described by means of specific embodiments, examples and applications thereof, numerous modifications and variations could be made thereto by those skilled in the art without departing from the scope of the invention set forth in the claims.

1. A human-computer interface system including:
- a sensor configured to transduce the location of a finger of a hand of a user;
- an exoskeleton configured to exchange mechanical energy with the finger; and
- an interface garment, including:
  - an interface laminate coupled to a counterpressure assembly, configured to stimulate the user by applying a pressure to the finger;
  - a first plurality of tactile actuators configured to apply a pressure to a finger of the user; and
  - a second plurality of tactile actuators configured to apply a pressure to a palm of the hand of the user.

2. The human-computer interface system of claim **1** wherein the sensor comprises at least one of: a magnetic sensor and an optical sensor.

3. The human-computer interface system of claim **2** further comprising a magnetic emitter coupled to the hand of the user.

4. The human-computer interface system of claim **1** wherein the interface laminate comprises a portion with at least 1 tactile actuator per square centimeter.

5. The human-computer interface system of claim **1** wherein the interface laminate comprises a tactile actuator with a displacement of at least 1 mm.

6. The human-computer interface system of claim **1** further comprising an intermediate layer between the interface laminate and skin of the user.

7. The human-computer interface system of claim 1 wherein the interface laminate is coupled to a vibration actuator.

8. The human-computer interface system of claim 7 wherein the vibration actuator comprises one of a piezoelectric actuator and a fluidic actuator.

9. The human-computer interface system of claim 1 wherein the counterpressure assembly comprises an armature and a tensile member.

10. The human-computer interface system of claim 9 wherein said tensile member comprises an elastic element.

11. The human-computer interface system of claim 9 wherein said armature contacts palmar and dorsal faces of a distal phalange of the finger, but does not contact medial and lateral faces of said finger.

12. The human-computer interface system of claim 1 wherein the interface garment comprises a palm assembly, including at least one of: a thenar assembly, a hypothenar assembly, and an interdigital assembly.

13. The human-computer interface system of claim 12 wherein the stiffness of said palm assembly varies between a first and second point along its surface.

14. The human-computer interface system of claim 13 having a less stiff portion adjacent to at least one of: a palmar interdigital crease, a transverse crease, and a thenar crease.

15. The human-computer interface system of claim 12 wherein the thenar assembly comprises a portion having a radius of curvature between 10 mm and 20 mm.

16. The human-computer interface system of claim 12 wherein the interdigital assembly comprises a portion having a radius of curvature between 56 mm and 94 mm.

17. The human-computer interface system of claim 1 wherein the interface garment comprises a wrist strap.

18. The human-computer interface system of claim 1 wherein the counterpressure assembly comprises a fastener configured to adjust pressure applied by the counterpressure assembly to a portion of the hand of the user.

19. The human-computer interface system of claim 1 comprising at least one of: an opisthenar plate, and a thenar plate.

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